



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-19403

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Fri, Jul 19, 2024 at 12:17 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

19-Jul-2024

Dear Mr. Khare:

The manuscript entitled "Optimized Data-Flow Integrity for Modern Compilers" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access
ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Optimized Data-Flow Integrity for Modern Compilers

AUTHORS: IRENE, DÍEZ-FRANCO; Ugarte-Pedrero, Xabier; Bringas, Pablo G.

ABSTRACT: Non-control-data attacks are those attacks that purely target and modify the non-control data of a program, such as boolean values, user input or configuration parameters, and leave the control flow of a program untouched. These attacks were considered a niche due to the high difficulty in crafting attacks that do not modify the control flow. However, in recent years researchers have already demonstrated that non-control-data attacks can be automatically constructed and that they pose a significant threat because they can compromise critical and widely used software, such as web browsers and the Linux kernel. Moreover, they can also be used to disable or bypass state-of-the-art software security techniques, such as control-flow integrity. The most promising technique to protect against non-control-data attacks is data-flow integrity, however, modern compilers do not implement this protection yet. In this work we present an optimized data-flow integrity implementation for modern compilers that reduces the amount of basic blocks that need to be protected, it also has broader security guarantees due to its more precise static analysis. Finally, we evaluate the completeness of our optimized data-flow integrity implementation.

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